

APOLLO HOSPITALS GROUP

Leads Flow Dashboard

Power BI Analytics Project Report

A comprehensive business intelligence solution delivering real-time insights on patient flow, bed occupancy, billing performance, and clinical service quality across Apollo Hospitals' operations.

KEY METRICS AT A GLANCE

Total Revenue	Admit Date	Discharge Date	Follow-up Date
Rs.190.43M	05-12-2022	12-01-2023	15-01-2023



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EXECUTIVE SUMMARY

The Apollo Hospitals Leads Flow Dashboard is a production-grade Power BI analytics solution designed to address one of the most pressing operational challenges in modern healthcare: the ability to convert raw hospital data into actionable intelligence. Built for Apollo Hospitals — one of India's largest and most recognised healthcare networks — this dashboard empowers hospital administrators, clinical directors, and operational managers to make evidence-based decisions in real time.

Hospitals today manage an overwhelming volume of patient touchpoints: admissions, discharge cycles, bed allocation, clinical diagnoses, billing transactions, and physician performance evaluations. Without a centralised analytics layer, these data streams exist in silos, limiting the organisation's ability to respond proactively to resource shortages, revenue leakage, or service quality decline.

Business Impact

By consolidating critical hospital KPIs — including a Total Bill Amount of Rs.190.43 Million, bed utilisation rates across Private, General, and ICU categories, and diagnosis-wise patient distribution — the dashboard enables Apollo Hospitals to reduce operational inefficiencies, optimise bed planning, and improve patient satisfaction scores.

This report documents the methodology, analytical findings, visual architecture, and strategic recommendations derived from the dashboard. It is intended to serve both as a technical portfolio deliverable and a strategic analytics narrative for stakeholders evaluating the organisation's data maturity.

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BUSINESS PROBLEM STATEMENT

Apollo Hospitals, like all large healthcare institutions, operates across multiple complex and interdependent dimensions simultaneously. The absence of a unified operational dashboard means that critical decision-making is often reactive rather than proactive. The following pain points formed the foundation of this project:

Problem Area	Operational Impact
Resource Underutilisation	Bed occupancy patterns were not consistently monitored, resulting in inefficient allocation across Private, General, and ICU wards.
Revenue Leakage & Billing Gaps	Fluctuating billing amounts across patient cohorts signalled untracked anomalies and potential revenue leakage with no analytical visibility.
Reactive Clinical Planning	Diagnosis-wise patient surges (e.g., viral infection seasons) were managed reactively, not forecasted, leading to capacity constraints.
Doctor Performance Blindspots	Physician feedback data existed but was not aggregated or visualised, preventing systematic service quality evaluation.
Fragmented Patient Journey Tracking	Admit, discharge, and follow-up timelines were not connected analytically, creating gaps in patient flow management and care continuity.

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PROJECT OBJECTIVES

01	Centralise Hospital KPIs Aggregate all critical metrics — billing, occupancy, patient flow, and feedback — into a single, unified dashboard interface.
02	Enable Bed Occupancy Intelligence Provide granular occupancy analysis by bed category (Private, General, ICU) to guide daily and strategic resource planning.
03	Track Diagnosis Trends Visualise patient volume by diagnosis type to support seasonal preparation and clinical department planning.
04	Revenue Monitoring & Trend Analysis Monitor total billing and identify fluctuation patterns that may indicate billing anomalies or service demand shifts.

05	Doctor Feedback Evaluation Quantify feedback volume per physician to drive performance management and quality improvement initiatives.
06	Flexible Filtering & Drill-Down Empower analysts with Date Range and Patient ID filters for granular, on-demand data exploration.

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DATASET OVERVIEW

The dataset underpinning this dashboard was sourced from Apollo Hospitals' operational records, structured as a flat-file CSV/Excel format. The raw data underwent a comprehensive data preparation pipeline within Power Query, ensuring analytical integrity before model construction.

4.1 Dataset Schema & Field Definitions

Patient ID	Unique alphanumeric identifier for each patient record
Admit Date	Date of patient admission (earliest: 05-Dec-2022)
Discharge Date	Date of patient discharge (latest: 12-Jan-2023)
Follow-up Date	Scheduled post-discharge follow-up date
Diagnosis	Primary clinical diagnosis (Viral Infection, Flu, Malaria, Typhoid, Pneumonia)
Bed Type	Ward category assigned: Private, General, or ICU
Doctor Name	Attending physician — linked to feedback metrics
Feedback Score	Patient-submitted service rating per consultation
Bill Amount (INR)	Total billing value per patient episode
Health Insurance	Whether the patient hold health insurance (Yes / No)

4.2 Data Preparation & Quality Assurance

- Removed duplicate Patient IDs and null billing records via Power Query.
- Standardised date formats across Admit, Discharge, and Follow-up columns.
- Replaced inconsistent diagnosis labels with normalised category names.
- Created calculated date-difference columns for Length of Stay (LOS) metrics.
- Validated bed-type categories and corrected miscoded ward entries.
- Applied conditional column logic to flag high-value billing outliers.

4.3 Data Summary

Total Patient Records	Aggregated across study period (Dec 2022 – Jan 2023)
Total Billing Revenue	Rs. 190.43 Million (INR)
Study Period	05-Dec-2022 to 15-Jan-2023 (incl. follow-ups)
Diagnosis Categories	5 (Viral Infection, Flu, Malaria, Typhoid, Pneumonia)
Bed Categories	3 (Private, General, ICU)

Analytical Tool	Power BI Desktop Power Query DAX
File Format	Excel / CSV (flat-file input)

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DASHBOARD OVERVIEW

The Leads Flow Dashboard was designed following enterprise UX principles — prioritising clarity, information hierarchy, and visual consistency. Built within Power BI Desktop, the dashboard delivers a single-screen operational command centre supported by interactive cross-filtering capabilities.

5.1 KPI Cards — Executive Metrics Panel

Four high-visibility KPI cards anchor the top of the dashboard, providing immediate visibility into the most business-critical metrics:



5.2 Bed Occupancy Analysis — Clustered Bar Chart

A clustered bar chart segments total bed utilisation across Private, General, and ICU categories. This visual enables ward managers to immediately identify pressure points in capacity planning. Private beds emerge as the dominant category — a finding that carries significant operational and revenue implications. The visual supports colour-encoded differentiation for rapid cognitive processing.

5.3 Doctor Feedback Volume — Horizontal Bar Chart

This visualisation aggregates feedback submission counts per attending physician. Rather than a simplistic rating display, feedback volume itself is a proxy for patient engagement and service interaction frequency. Physicians with disproportionately low feedback counts are flagged for further quality review. This supports HR and clinical governance processes.

5.4 Diagnosis-Wise Patient Count — Donut / Bar Chart

A diagnosis breakdown visual maps patient volume across all five clinical categories: Viral Infection, Flu, Malaria, Typhoid, and Pneumonia. The distribution immediately reveals the dominance of Viral Infection as the leading admissions driver, informing both clinical stocking and physician scheduling decisions.

5.5 Billing Amount Trend — Line / Area Chart

A time-series visualisation tracks total billing amounts across the study period. The trend line exposes significant billing volatility — peaks aligning with high-admission periods and troughs correlating with holiday or discharge-heavy windows. This visual is essential for finance teams monitoring revenue cycle health.

5.6 Interactive Filters — Slicers

Two dynamic slicers empower users to drill into specific data segments: (1) Date Range Slicer — a calendar-based filter enabling period-specific analysis for billing cycles, seasonal demand, or audit windows. (2)

Patient ID Filter — a search-enabled slicer for individual patient-level investigation, supporting both clinical and administrative use cases.

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DATA ANALYSIS & KEY INSIGHTS

Beyond surface-level data visualisation, the true value of this project lies in the depth of analytical interpretation. The following insights represent findings derived from cross-visual analysis, DAX-driven calculations, and domain-specific healthcare knowledge.

6.1 Viral Infection Dominates Admissions

Viral infections account for the largest share of patient admissions across the study period. This is consistent with seasonal epidemiological patterns in the Indian subcontinent, where respiratory and vector-borne viral illnesses spike during the post-monsoon winter transition (November–January).

Key Insight

Viral infection patients are high-frequency, moderate-acuity cases — driving admissions volume without proportionately increasing ICU utilisation. This creates a specific operational profile: high General and Private ward demand, rapid turnover cycles, and elevated pharmacy & consumables expenditure.

The implication for Apollo is clear: seasonal viral surge periods should trigger pre-emptive capacity protocols — including surge bed activation, accelerated discharge planning, and pharmacy stock pre-loading — rather than reactive responses.

6.2 Private Bed Occupancy — The Revenue Driver

Private bed occupancy consistently outpaces both General and ICU categories. This finding reflects a well-documented trend in Indian private healthcare: a growing middle-class demographic with insurance coverage or out-of-pocket spending capacity actively seeks premium ward facilities.

Bed Type	Occupancy Level	Operational Characteristics
Private	Highest	Premium pricing, preferred by insured patients, longest average LOS
General	Moderate	High volume, lower revenue per bed, rapid turnover
ICU	Variable	Low absolute occupancy but highest revenue per patient day

Business Impact

Private bed revenue concentration creates a dependency risk. A shift in insurance reimbursement policies or a macro-economic downturn could significantly impact Apollo's revenue baseline. Portfolio diversification — through tiered General ward upgrades — is recommended.

6.3 Billing Amount Volatility — Revenue Cycle Implications

The billing trend analysis reveals pronounced fluctuations across the 40-day study window. Peaks in billing correlate with high-admission periods (week 1–2 of December), while billing troughs align with the Christmas–New Year holiday window, during which elective procedures typically decline.

These fluctuations, while partially seasonal, may also mask billing process inefficiencies — delayed coding, late insurance submissions, or discharge-billing lag. A structured billing audit for gap weeks is strongly recommended to ensure revenue integrity within each 30-day financial cycle.

6.4 Doctor Feedback — Quality Signals

Feedback volume analysis reveals uneven distribution across the physician panel. Senior specialists attract substantially higher feedback submission rates — a function of both patient volume and interaction depth. Notably, low-feedback physicians are not necessarily underperforming; they may be managing lower-contact specialties or operating in procedural roles with limited patient communication time.

Key Insight

Feedback volume is a leading indicator of patient engagement, not a standalone performance metric. When cross-referenced with diagnosis type and LOS, it provides a richer picture of clinical interaction quality and service consistency.

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BUSINESS RECOMMENDATIONS

The following recommendations are grounded in the analytical insights derived from the dashboard and are structured for immediate, medium-term, and strategic implementation:

IMMEDIATE	Implement Surge Capacity Protocols Activate pre-defined surge protocols for Private and General wards during viral infection season (Oct–Feb). Pre-position nursing staff, pharmacy stock, and diagnostic equipment at 120% capacity thresholds.
IMMEDIATE	Revenue Cycle Audit — Billing Gaps Commission a targeted billing audit for holiday-period revenue troughs to identify and recover delayed billing submissions, uncoded procedures, and insurance claim backlogs.
MEDIUM-TERM	ICU Capacity Planning & Predictive Modelling Develop a 30-day ICU demand forecast model using historical diagnosis trends, LOS patterns, and seasonal indicators. This will reduce unplanned ICU overflow events and improve critical care resource allocation.
MEDIUM-TERM	Doctor Performance Framework Integrate feedback volume, patient satisfaction scores, and diagnosis-outcome data into a holistic physician performance scorecard reviewed quarterly by clinical governance committees.
STRATEGIC	Predictive Healthcare Analytics Platform Evolve the current descriptive dashboard into a predictive analytics platform using machine learning models (Python/Azure ML) to forecast admission surges, bed demand, and billing anomalies 14–30 days in advance.
STRATEGIC	General Ward Revenue Optimisation Explore tiered General ward upgrades (semi-private modules) to reduce Private bed revenue concentration risk while capturing mid-market patient segments willing to pay a premium over standard General ward rates.

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SKILLS DEMONSTRATED

This project demonstrates a comprehensive set of technical and analytical competencies relevant to modern Data Analyst and Business Intelligence roles:

Power BI Development	End-to-end dashboard development including data modelling, relationships, measures, and report-layer design. Proficient in Power BI Desktop, Report View, Model View, and DAX Studio integration.
DAX (Data Analysis Expressions)	Authored advanced DAX measures for KPI calculations, time-intelligence functions (DATEADD, TOTALYTD), CALCULATE with complex filter contexts, and dynamic ranking measures using RANKX.
Power Query (M Language)	Data ingestion, transformation, and quality assurance pipeline built entirely within Power Query — including conditional columns, merge queries, unpivoting, and custom M functions.
Healthcare Analytics	Domain expertise in hospital operational metrics — bed occupancy, patient flow, LOS analysis, diagnosis categorisation, and clinical performance indicators — applied within a real-world hospital context.
Data Visualisation & UX Design	Designed a visually consistent, user-centric dashboard layout adhering to corporate colour standards, accessibility contrast principles, and information hierarchy best practices.
Business Intelligence & Storytelling	Translated raw operational data into executive-ready insights with clear business narratives, actionable recommendations, and stakeholder-appropriate communication formats.
Excel & Data Preparation	Proficient in Excel-based data pre-processing: pivot tables, VLOOKUP/INDEX-MATCH, data validation, and structured table formatting as input for BI tools.
Analytical Thinking	Ability to identify patterns, anomalies, and business-critical signals from multi-dimensional healthcare datasets and translate findings into operational and strategic recommendations.

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FUTURE SCOPE

The current dashboard represents Phase 1 of a broader healthcare analytics roadmap. The following capabilities are planned for subsequent development phases, aligned with Apollo Hospitals' digital transformation agenda:

[Phase 2] AI & Machine Learning Integration

Embed predictive models (scikit-learn, Azure ML) to forecast bed demand, billing anomaly detection, and readmission risk scoring directly within the Power BI environment via Python visuals.

[Phase 2] Real-Time Data Streaming

Replace static CSV imports with live data connections using DirectQuery to hospital SQL databases, enabling real-time dashboard refresh for intraday operational monitoring.

[Phase 3] SQL & Data Warehouse Integration

Migrate the data model to a structured SQL data warehouse (Azure Synapse or PostgreSQL), enabling historical multi-year analysis, complex joins across clinical systems, and enterprise-scale reporting.

[Phase 3] Cloud Deployment & Power BI Service

Publish the dashboard to Power BI Service with row-level security (RLS), scheduled refresh, and role-based access control for deployment across Apollo's hospital network.

[Phase 4] NLP Patient Feedback Analysis

Apply natural language processing (Azure Cognitive Services or HuggingFace) to analyse free-text patient feedback, extracting sentiment scores and theme clusters for clinical quality improvement.

[Phase 4] Multi-Hospital Network Dashboard

Scale the solution to a network-level analytics platform covering all Apollo Hospital branches — enabling cross-facility benchmarking, centralised KPI governance, and group-level executive reporting.

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CONCLUSION

The Apollo Hospitals Leads Flow Dashboard is more than a data visualisation project — it is a demonstration of how business intelligence, applied with domain knowledge and analytical rigour, can fundamentally transform how healthcare organisations operate and make decisions.

By designing a dashboard that speaks to multiple organisational stakeholders — clinical directors, operational managers, finance teams, and hospital administrators — this project illustrates the core value proposition of a skilled Data Analyst: the ability to bridge the gap between raw data and real decisions.

The insights generated — from the dominance of viral infection admissions and the Private bed revenue concentration, to billing volatility and physician feedback patterns — are not merely academic observations. They are operational signals that, when acted upon, translate directly into improved patient outcomes, optimised resource deployment, and stronger financial performance.

Portfolio Statement

This project reflects my capability to independently conceive, build, and interpret a complete analytics solution — from raw data ingestion and Power Query transformation, through DAX-driven KPI modelling, to professional dashboard design and strategic business communication. I bring both the technical depth and the business acumen to deliver

I am actively seeking roles where I can apply these skills to complex, high-impact analytics challenges. I welcome the opportunity to discuss this project and my broader portfolio with prospective employers, clients, and collaborators.

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